

# Marginal Matching Does Not License Factorized Sampling: Auditing Conditional Style Leakage in Factorized Generative Models

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## Abstract

Factorized generative models regularize a style latent  $z_s$  toward a fixed prior using a marginal statistic,  $q(z_s) \approx \mathcal{N}(0, I)$ , and treat this as a certificate that  $z_s$  is class-invariant. We show the certificate is invalid: matching the marginal places no constraint on the class conditionals  $q(z_s | y=k)$ , so  $z_s$  can be exactly Gaussian in aggregate while remaining maximally informative about the class label  $y$ . We call the resulting failure *conditional style leakage* and give a diagnostic for it based on inter-class style separation and linear probing. Across four unsupervised baselines (VAE, WAE-MMD,  $\beta$ -TCVAE, FactorVAE) and a case-study model, F-CS-WAE, every model with near-zero global MMD nonetheless allows a linear probe to recover  $y$  from  $z_s$  with 74–100% accuracy (chance 10%); on MNIST, F-CS-WAE reaches 99.15% clustering accuracy while class-conditional generation succeeds only 15% of the time under the sampling procedure that marginal matching is supposed to license. No single one of six one-at-a-time perturbations (latent capacity, training curriculum, prior geometry, supervision) eliminates the leakage on either MNIST or CIFAR-10, so it is not attributable to any one of these design choices alone – though on CIFAR-10 its magnitude is highly sensitive to capacity, which a MNIST-only check would have missed. We test two remedies. A supervised per-class style MMD substantially reduces, but does not eliminate, the leakage, at a clustering cost that is dataset-dependent rather than a fixed price of the fix. A post-hoc class-conditional style prior fully repairs MNIST generation without retraining, but only partially repairs CIFAR-10, so a remedy validated on one dataset does not automatically generalize to another. The broader point is methodological: no divergence computed on  $q(z_s)$  alone can certify  $z_s \perp y$ , and factorized generative models that report only marginal statistics have not actually verified the property they claim.

## 1 Introduction

Matching the aggregate style posterior to a Gaussian prior is a statement about a marginal distribution; class-invariant style is a statement about a family of conditional distributions. The two are not the same claim, and satisfying the first does not satisfy the second:  $q(z_s)$  can be exactly  $\mathcal{N}(0, I)$  while

each  $q(z_s | y=k)$  occupies a different region of latent space, so long as those regions average out to the right aggregate shape. A regularizer that only ever looks at  $q(z_s)$  has no way to see this, and therefore no way to prevent it.

**Where this bites.** Consider a toy version of the standard factorized setup: a semantic variable  $z_c$  carries class identity, a style variable  $z_s$  is meant to carry everything else, and  $z_s$  is trained to satisfy  $q(z_s) \approx \mathcal{N}(0, I)$ . During training, the decoder only ever sees  $(z_c, z_s)$  pairs drawn from the same image, so if  $q(z_s | y=k)$  happens to differ across classes—which nothing in the training objective prevents—the decoder is free to learn class-specific interactions between  $z_c$  and  $z_s$ . At generation time, a class prior supplies  $z_c$  for class  $k$  while  $z_s$  is drawn from the *marginal*,  $\mathcal{N}(0, I)$ , which averages over all classes. The resulting pair  $(z_c, z_s)$  is one the decoder never saw during training, and its output is unconstrained by anything the model learned. We call this failure *conditional style leakage*.

**Why the usual metrics do not catch it.** The failure is silent precisely because it is invisible to the metrics a practitioner would normally check. Reconstruction quality is measured on the posterior manifold –  $z_s$  inferred from a real image – where the class-specific interaction the decoder learned is exactly the one being exercised, so reconstruction looks fine. Global MMD is a marginal statistic by definition and, as above, is compatible with arbitrary conditional structure. Clustering accuracy depends only on the semantic variable  $z_c$  and says nothing about  $z_s$  at all. A model can therefore score well on every diagnostic a typical evaluation protocol reports and still fail completely at class-conditional generation, which is the one capability the factorization exists to provide.

**A diagnostic, not just an example.** Because the failure does not show up in standard metrics, detecting it requires measurements taken directly on the conditional structure of  $z_s$ : we use inter-class style separation  $\Delta_{\text{inter}}$  and linear probing accuracy of  $y$  from  $z_s$  (Section 3.5, which also defines a complementary kernel-based diagnostic,  $\text{HSIC}(z_s, y)$ ), alongside the marginal MMD that current practice already reports. A model with near-zero MMD but large  $\Delta_{\text{inter}}$  or elevated

linear-probe accuracy is leaking class information through a channel that was supposed to be class-invariant.

**What we find.** Applying this diagnostic across five model families on MNIST, every model with near-zero global MMD still allows a linear probe to recover the class label from  $z_s$  with 74–100% accuracy against a 10% chance rate (Section 6.1); this includes  $\beta$ -TCVAE and FactorVAE, which are explicitly designed to penalize latent dependence. As a concentrated case study, we instantiate F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder), a model that separates a hyperspherical semantic variable from a Euclidean style variable; without any conditional regularization it reaches 99.15% clustering accuracy on MNIST while class-conditional generation succeeds only 15% of the time. We then test two remedies – a supervised per-class style MMD applied at training time, and a post-hoc class-conditional style prior applied at sampling time – and report where each helps, where it does not fully fix the problem, and where a fix validated on MNIST fails to transfer to CIFAR-10 (Section 6.6).

**Scope.** We use F-CS-WAE to exhibit and repair the failure, not to argue it is a better model than the baselines it is compared against: the cross-model comparison in Section 6.1 does not control for encoder capacity or training budget, the CIFAR-10 baseline comparison in Section 5 is single-seed, and the proposed per-class style MMD is a supervised, label-guided regularizer, not an unsupervised disentanglement method. These scope limits are stated where they arise and collected in Section 7.1.

**Contributions.** Our contributions are fourfold. **(1) A decomposition of factorized-sampling validity.** We show the mismatch a factorized sampler incurs decomposes exactly into four conceptually distinct, non-negative terms (Theorem 1): global style-prior mismatch, conditional style leakage, semantic-prior mismatch, and within-class semantic-style dependence. Two corollaries make the consequence precise: valid factorized sampling requires all four terms to vanish (so class-invariant style is necessary but not sufficient, Corollary 1), and the latent-space mismatch upper-bounds the resulting image-space mismatch but not conversely (Corollary 2), which is what makes a decoder-level intervention necessary rather than redundant. **(2) A conditional audit protocol.** We build a lightweight suite of finite-sample proxies for the four terms and apply it across five model families, showing the conditional leakage the decomposition predicts is real even when – for three of the five – it is invisible to a naive 2D projection of the latent (Section 6.1, Appendix D). **(3) A decoder-sensitivity and robustness study.** A controlled latent-swap intervention shows the decoder is measurably more sensitive to the leaked style code than to the semantic code, and that this sensitivity shifts

once the remedy is applied (Section 6.3); a six-way, one-at-a-time perturbation of capacity, curriculum, prior geometry, and supervision on both datasets shows no single design choice eliminates the leakage, while revealing that its magnitude is strongly capacity-sensitive on CIFAR-10 in a way a MNIST-only check would miss (Section 6.8). **(4) An evaluation of representation- and sampling-level repairs, with a negative result.** We study per-class style MMD – itself a direct, conditional-matching instantiation of the variational-fair-autoencoder idea [19] applied to a style code, not a new mechanism – and a post-hoc class-conditional prior, and report where each helps, where it does not, and a negative transfer result: the sampling recipe that fully repairs MNIST generation only partially repairs CIFAR-10 (Sections 4, 6.7, 6.6).

## 2 Related Work

We organize prior work around a single question: does the method’s training or evaluation protocol certify conditional style invariance,  $z_s \perp y$ , or only a marginal or joint statistic that is compatible with  $z_s \not\perp y$ ?

**Variational and Wasserstein auto-encoders.** VAEs [13] impose a per-example KL penalty toward a fixed Gaussian prior; WAEs [23] relax this to a divergence between the *aggregated* posterior and the prior, computed with kernel two-sample tests [5] or projected distances [15]. The VAE penalty is not purely an aggregate regularizer in disguise: averaged over the data distribution it decomposes exactly as  $\mathbb{E}_{q(x)}[\text{KL}(q(z|x) \| p(z))] = I_q(X; Z) + \text{KL}(q(z) \| p(z))$  [9], an information-capacity term plus an aggregate-prior-mismatch term structurally identical to what WAE-MMD targets directly. Neither term, however, certifies  $I(z_s; y) = 0$ : the mutual-information term bounds how much  $Z$  encodes about  $X$  in general, not specifically about the label  $y$ , and the aggregate term is exactly the marginal statistic Proposition 1 shows is insufficient. WAE-MMD matches the aggregate posterior only; VAE’s per-example KL is an information-bottleneck term plus that same aggregate match;  $\beta$ -TCVAE/FactorVAE below penalize dependence *among latent dimensions*, a third, still-different target. None of these three families evaluates  $q(z_s | y)$ , so none certifies conditional invariance even when its own target is well matched.

**Disentanglement and total correlation.**  $\beta$ -VAE [8] increases the KL weight to encourage disentangled factors;  $\beta$ -TCVAE [1] and FactorVAE [12] isolate and penalize the total correlation between latent dimensions directly. These methods get closer to our concern, since total correlation is a joint-distribution statistic rather than a purely marginal one, but the joint they penalize is over *latent dimensions*, not over the latent and the label. Table 2 shows both still leak:  $\beta$ -

TCVAE and FactorVAE have among the lowest global MMD in our comparison, yet a linear probe recovers the class label from their style latent at 73.9% and 75.9% accuracy respectively, far above the 10% chance rate.

**Conditional and semi-supervised generative models.** Semi-supervised VAEs [14] and CVAE [22] condition generation on the label directly, which sidesteps needing  $z_c$  to encode class identity on its own, but neither checks whether the *residual* latent still carries label information after conditioning. StyleGAN [11] factorizes coarse structure from fine style via a mapping network in a class-unconditional regime, so the question does not arise there in the same form. Our setting requires both a class-conditional generator and a label-free clustering signal from the same model, which is what makes residual leakage in  $z_s$  consequential rather than incidental.

**Deep generative clustering.** DEC [25], IDEC [6], and VaDE [10] evaluate almost exclusively via ACC/NMI/ARI on the semantic or cluster-assignment variable. This protocol cannot see conditional style leakage by construction, since it never inspects  $z_s$ : our MNIST case study makes this concrete, reaching 99.15% clustering accuracy (Section 6.1) on exactly the checkpoint whose class-conditional generation succeeds only 15% of the time (Section 6.2).

**Invariant and nuisance-free representations.** Outside the generative-modeling literature, several methods target exactly the dependency this paper studies,  $z_s \not\perp y$ , through mechanisms other than an aggregate divergence. The variational fair autoencoder [19] matches a nuisance-conditional posterior toward a common target per nuisance value – the per-class style MMD we use in Section 4.2 is a direct instantiation of this idea applied to a style code, not a new mechanism. Gradient-reversal domain adaptation [3] and Fader Networks [17] instead remove label information from a representation adversarially, in image or feature space; Moyer et al. [20] achieve a similar effect without an adversary via a different information-theoretic penalty. None of these methods evaluates class-conditional generation under a factorized sampler of the kind Section 3 analyzes, so none of them establishes whether removing  $z_s \not\perp y$  alone is sufficient to make independent sampling reliable – the question Section 3.3 shows it is not, since it leaves within-class  $z_c$ - $z_s$  dependence untouched.

**Hyperspherical latent variables.** S-VAE [2] replaces the Gaussian posterior with a von Mises-Fisher distribution; Spherical Cauchy VAEs [21] give a heavier-tailed alternative with a Möbius-transform reparameterization. We use the latter for the semantic variable in our case-study model (Section 4.3). This choice of geometry is an implementation

detail of the case study, not a claim about the leakage problem, which is defined independently of how  $z_c$  is parameterized.

### 3 The Conditional Style Leakage Problem

#### 3.1 Setting and Notation

Let  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  be a labeled dataset with  $K$  classes. A factorized model encodes each input into a semantic variable  $z_c$  and a style variable  $z_s$ . The style variable is regularized by

$$\mathcal{L}_{\text{style}} = D(q(z_s), \mathcal{N}(0, I)), \quad (1)$$

where  $D$  is any aggregate divergence and  $q(z_s) = \mathbb{E}_y[q(z_s | y)]$ . Class-conditional generation samples  $z_s \sim \mathcal{N}(0, I)$  independently of  $z_c$ .

#### 3.2 Formal Gap

**Proposition 1** (Marginal matching does not certify conditional independence). *Let  $z_s \sim \mathcal{N}(0, I)$  on  $\mathbb{R}^{d_s}$ , and let  $\{A_k\}_{k=1}^K$  be any measurable partition of  $\mathbb{R}^{d_s}$  with  $P(z_s \in A_k) = 1/K$  for each  $k$  (such a partition always exists, e.g. via level sets of a continuous statistic of  $z_s$ ). Define the label  $y$  as the deterministic function  $y = k \iff z_s \in A_k$ . Then:*

- (i) *the marginal is unchanged,  $q(z_s) = \mathcal{N}(0, I)$  exactly, so  $D(q(z_s), \mathcal{N}(0, I)) = 0$  for every divergence  $D$ ; and*
- (ii)  *$I(z_s; y) = H(y) = \log K$ , the maximum mutual information achievable between  $z_s$  and a  $K$ -ary label.*

*Proof.* (i)  $y$  is defined post hoc as a measurable function of  $z_s$ ; the distribution of  $z_s$  itself is never altered, so  $q(z_s) = \mathcal{N}(0, I)$  holds exactly and every divergence to  $\mathcal{N}(0, I)$ , aggregate MMD included, is identically zero. (ii) Since  $y$  is a deterministic function of  $z_s$ ,  $H(y | z_s) = 0$ , so  $I(z_s; y) = H(y) - H(y | z_s) = H(y)$ . By construction  $P(y = k) = 1/K$  for all  $k$ , so  $H(y) = \log K$ , which is also the maximum possible entropy – and hence the maximum possible mutual information with any other variable – for a  $K$ -ary label.  $\square$

**Remark 1.** *The construction is deliberately adversarial:  $\{A_k\}$  need not resemble anything a trained decoder would produce, and  $q(z_s | y=k)$  here is  $z_s$  restricted to  $A_k$  and renormalized, not a smooth family with class-specific means. That is the point – it shows that no divergence computed on  $q(z_s)$  alone, MMD or otherwise, can rule out  $z_s$  being maximally class-informative, regardless of how the dependence is structured geometrically. This is a no-certificate result: it does not say marginal matching typically leaks this badly, only that no marginal statistic, however small, can rule the possibility out. Section 3.3 complements this worst-case result with a decomposition of the mismatch a factorized sampler incurs in general, and Section 3.5 discusses the finite-sample proxies, including  $\Delta_{\text{inter}}$ , used to detect it in practice.*

### 3.3 Full Class-Conditional Mismatch Decomposition

Proposition 1 is the extremal case of term (2) below: it exhibits a  $q(z_s)$  for which term (1) is exactly zero while term (2) is maximal, showing the gap between them admits no universal bound. The following theorem decomposes the mismatch a factorized sampler actually incurs, for any encoder  $q$  – not only that adversarial construction – into four named, non-negative terms.

**Theorem 1** (Factorized-sampling mismatch decomposition). *Let  $y$  be uniform on  $\{1, \dots, K\}$ , let  $q(z_c, z_s \mid y=k)$  be the encoder’s joint class conditional, let  $p(z_c \mid y=k)$  and  $p(z_s)$  be the semantic and style priors (Section 3), and let  $q(z_s) = \mathbb{E}_y[q(z_s \mid y)]$  be the aggregate style posterior (Eq. 1). Define the factorized-sampling mismatch*

$$M_{\text{fact}} := \mathbb{E}_y \left[ \text{KL}(q(z_c, z_s \mid y) \parallel p(z_c \mid y) p(z_s)) \right].$$

Then

$$M_{\text{fact}} = I_q(z_c; z_s \mid y) \quad (\text{term 4})$$

$$+ \mathbb{E}_y [\text{KL}(q(z_c \mid y) \parallel p(z_c \mid y))] \quad (\text{term 3})$$

$$+ I_q(z_s; y) \quad (\text{term 2})$$

$$+ \text{KL}(q(z_s) \parallel p(z_s)). \quad (\text{term 1})$$

All four terms are non-negative, so each individually lower-bounds  $M_{\text{fact}}$ .

*Proof.* Fix  $k$  and expand the class- $k$  term of  $M_{\text{fact}}$  by inserting  $\pm \log q(z_c \mid k) \pm \log q(z_s \mid k)$  into the integrand:

$$\begin{aligned} & \text{KL}(q(z_c, z_s \mid k) \parallel p(z_c \mid k) p(z_s)) \\ &= \mathbb{E}_{q(z_c, z_s \mid k)} \left[ \log \frac{q(z_c, z_s \mid k)}{q(z_c \mid k) q(z_s \mid k)} \right] \\ &+ \mathbb{E}_{q(z_c \mid k)} \left[ \log \frac{q(z_c \mid k)}{p(z_c \mid k)} \right] \\ &+ \mathbb{E}_{q(z_s \mid k)} \left[ \log \frac{q(z_s \mid k)}{p(z_s)} \right] \\ &= I_q(z_c; z_s \mid y=k) + \text{KL}(q(z_c \mid k) \parallel p(z_c \mid k)) \\ &+ \text{KL}(q(z_s \mid k) \parallel p(z_s)), \end{aligned}$$

where the first term is, by definition, the mutual information between  $z_c$  and  $z_s$  under  $q(\cdot, \cdot \mid y=k)$ . Taking  $\mathbb{E}_y[\cdot]$  gives terms (4) and (3) directly, since  $\mathbb{E}_y[I_q(z_c; z_s \mid y=k)] = I_q(z_c; z_s \mid y)$  by definition of conditional mutual information. For the remaining piece,  $\mathbb{E}_y[\text{KL}(q(z_s \mid y) \parallel p(z_s))]$ , insert  $\pm \log q(z_s)$  and use  $q(z_s) = \mathbb{E}_y[q(z_s \mid y)]$  (Section 3):

$$\begin{aligned} \mathbb{E}_y [\text{KL}(q(z_s \mid y) \parallel p(z_s))] &= \mathbb{E}_y [\underbrace{\text{KL}(q(z_s \mid y) \parallel q(z_s))}_{= I_q(z_s; y)} \\ &+ \underbrace{\text{KL}(q(z_s) \parallel p(z_s))}_{\text{term (1)}}], \end{aligned}$$

where the first bracket is exactly  $I_q(z_s; y)$  (the average KL of a conditional from its own marginal is, by definition, the mutual information). Non-negativity of every term follows from non-negativity of KL divergence and of mutual information (itself a KL divergence).  $\square$

Two corollaries connect the decomposition to what a practitioner actually cares about – valid sampling and faithful generation.

**Corollary 1** (Exact sampling validity). *The factorized sampler reproduces the encoder’s per-class joint,  $q(z_c, z_s \mid y=k) = p(z_c \mid y=k) p(z_s)$  for almost every  $k$ , if and only if  $M_{\text{fact}} = 0$ , i.e. if and only if all four terms of Theorem 1 vanish simultaneously. In particular, class-invariant style ( $I_q(z_s; y) = 0$ , term 2) is necessary but not sufficient for valid factorized sampling: it must be accompanied by a matched aggregate style prior (term 1), a matched semantic prior (term 3), and within-class independence of  $z_c$  and  $z_s$  (term 4).*

*Proof.* Every term of Theorem 1 is a non-negative expectation of a KL divergence (terms 2 and 4 are mutual informations, themselves KL divergences), so their sum  $M_{\text{fact}}$  is zero iff each term is zero; and  $\mathbb{E}_y[\text{KL}(\cdot \parallel \cdot)] = 0$  iff the integrand vanishes for a.e.  $y$ , iff the two per-class distributions coincide a.e.  $\square$

**Corollary 2** (Decoder pushforward bound). *Let  $\text{Dec}$  be the decoder and  $\text{Dec}_\# \mu$  the law of  $\text{Dec}(z_c, z_s)$  when  $(z_c, z_s) \sim \mu$ . The gap between the true class-conditional image distribution and the one the factorized sampler produces is bounded by the latent-space mismatch:*

$$\mathbb{E}_y [\text{KL}(\text{Dec}_\# q(\cdot \mid y) \parallel \text{Dec}_\# [p(z_c \mid y) p(z_s)])] \leq M_{\text{fact}}.$$

*Proof.* The data-processing inequality states that pushing two distributions through the same Markov kernel (here  $\text{Dec}$ ) cannot increase their KL divergence; applying it inside  $\mathbb{E}_y[\cdot]$  and invoking Theorem 1 gives the bound.  $\square$

**Remark 2.** *Corollary 2 runs one way only: a small  $M_{\text{fact}}$  suffices for faithful class-conditional generation, but a large  $M_{\text{fact}}$  does not force a visible failure, because a decoder can be insensitive to the mismatched latent directions and map distinct latent distributions to nearly identical image distributions. Whether the trained decoder actually conditions on the leaked, off-joint-support structure that terms (2) and (4) quantify is therefore an empirical question the decomposition does not settle – which is exactly what the latent-swap intervention of Section 6.3 measures, and why that experiment is not redundant with Theorem 1.*

Each term is a conceptually distinct failure mode with a distinct remedy, and each has a finite-sample proxy we can track – though we emphasize up front that we measure the proxies, not the exact information quantities, and never

estimate  $M_{\text{fact}}$  as a whole. Term (1),  $\text{KL}(q(z_s)||p(z_s))$ , is global style-prior mismatch – what  $\mathcal{L}_{\text{style}}$  (Eq. 1) targets, with global MMD as its proxy. Term (2),  $I_q(z_s; y)$ , is exactly the conditional style leakage this paper studies, targeted by the per-class style MMD remedy (Eq. 3) and proxied by  $\Delta_{\text{inter}}$  and LP (Section 3.5). Term (3),  $\mathbb{E}_y[\text{KL}(q(z_c | y)||p(z_c | y))]$ , is semantic-prior mismatch, targeted by  $\mathcal{L}_{\text{class}}$  (Appendix A). Term (4),  $I_q(z_c; z_s | y)$  – within-class dependence between the semantic and style codes – is targeted by *none* of the above: every existing loss term operates on  $z_c$  or  $z_s$  separately within a class, never on their joint. We introduce a JointMMD proxy for it below (Section 3.5) but treat that proxy as suggestive rather than conclusive, for the calibration reasons stated there. We state the decomposition in KL because the chain-rule identity is exact for KL; the case-study model’s losses use MMD as a differentiable, easier-to-estimate stand-in for the corresponding KL term (Eq. 1 already calls  $D$  “any aggregate divergence”), and the proxies above are correlates of, not unbiased estimators of, the terms they track.

### 3.4 The Decoder Amplification Mechanism

During training the decoder receives  $(z_c, z_s)$  pairs from the same image; both encode  $y$  implicitly,  $z_s$  because  $q(z_s | y=k)$  is not class-invariant. At generation time  $z_c \sim p(z_c | y=k)$  encodes a specific class while  $z_s \sim \mathcal{N}(0, I)$  averages over all classes, producing a cross-class pair never seen in training. Severity depends on how class-structured  $q(z_s | y)$  becomes: visually uniform classes (MNIST digits) force more class information into  $z_s$ , amplifying the failure; high-diversity classes (CIFAR-10) leave more class-neutral variation in  $z_s$ . Section 6.5 tests this prediction directly.

### 3.5 Diagnostic Metrics

**Global MMD (term 1).**  $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$ : near zero signals only that marginal matching succeeded.

**Inter-class style separation (term 2).**  $\Delta_{\text{inter}} = \frac{1}{\binom{K}{2}} \sum_{j < k} \|\bar{\mu}_s^{(j)} - \bar{\mu}_s^{(k)}\|_2$ ,  $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$ : large values indicate class-dependent style means despite small global MMD. This is a finite-sample, mean-separation statistic; it is a convenient, easily computed symptom of conditional dependence, not what Proposition 1 is about. That proposition concerns statistical dependence in general (formalized as mutual information) and holds even when class-conditional means coincide, so long as higher-order structure differs.

**Linear probing accuracy LP( $z_s \rightarrow y$ ) (term 2).** Accuracy of a linear classifier predicting  $y$  from  $z_s$ ; well above chance  $1/K$  confirms recoverable class information.

**HSIC( $z_s, y$ ) (term 2) – defined, not reported.** A kernel-based dependence measure [4] would in principle proxy term (2) as well; non-zero indicates  $z_s \not\perp y$  regardless of marginal shape. We define it for completeness but do *not* report it, because our fixed single-bandwidth estimator saturates at  $z_s$ ’s empirical scale (returning a near-constant value across checkpoints whose  $\Delta_{\text{inter}}$  and LP vary widely) – the same single-bandwidth failure mode the multi-scale MMD estimator is designed to avoid. A properly bandwidth-calibrated HSIC is left to future work; we rely on  $\Delta_{\text{inter}}$  and LP for term (2) throughout.

**Semantic-prior mismatch (term 3).** Already computed every training run as  $\mathcal{L}_{\text{class}}$  (Appendix A); no new computation is needed to report it as a diagnostic value alongside the others.

**Within-class  $z_c$ - $z_s$  dependence (term 4) – proposed proxy.** No existing diagnostic in this paper isolates  $I_q(z_c; z_s | y)$ . A natural finite-sample proxy is a *joint* MMD that compares real  $(z_c, z_s)$  pairs within a class against independently-recombined pairs from the same class:

$$\text{JointMMD} = \frac{1}{K} \sum_{k=1}^K \text{MMD}^2(\{(z_c^i, z_s^i)\}_{y_i=k}, \{(z_c^i, z_s^{\pi(i)})\}_{y_i=k}),$$

where  $\pi$  permutes style codes within class  $k$ , breaking exactly the within-class  $z_c$ - $z_s$  coupling term (4) measures while leaving both per-class marginals – and therefore terms (1)–(3) – unchanged. We report this proxy on the  $\delta=0/\delta=1$  checkpoints of both datasets (MNIST 0.00372  $\rightarrow$  0.00432; CIFAR-10 0.00359  $\rightarrow$  0.00435) but treat it as *suggestive, not conclusive*, and flag three calibration limits explicitly. (i) Each number is a single biased MMD estimate on one checkpoint with no error bars, permutation-null baseline, or seed replication, so we cannot attach significance to the small  $\delta=0$ -to-  $\delta=1$  change; the most we can say is that JointMMD shows *no observable reduction* under per-class style MMD, consistent with – but not sufficient to establish – the hypothesis that the remedy (which targets terms 1–2) leaves term (4) unaddressed (Section 6.4). (ii) The MMD magnitudes are *not* directly comparable across MNIST and CIFAR-10: without per-block feature normalization and a matched kernel bandwidth, equal MMD values on two different latent distributions do not imply equal underlying dependence, so we do not use the cross-dataset near-equality to draw conclusions about term (4)’s role in the dataset asymmetry (contrast the weaker claim we make in Section 6.6). A conclusive term-4 measurement would add a within-class permutation null distribution, a conditional-HSIC cross-check, bandwidth-sensitivity curves, and multi-seed error bars; we mark this as the single most important diagnostic gap in the audit suite (Section 7.1).

A model with near-zero global MMD but large  $\Delta_{\text{inter}}$  or elevated LP is exhibiting conditional style leakage.

## 4 Method

### 4.1 Problem Setting: Label-Guided Generative Clustering

Given labeled images  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  with  $K$  classes, we want (1) reconstruction that preserves visual content, (2) clustering that recovers class structure from semantic codes without using labels at evaluation time, and (3) class-conditional generation that produces class-consistent images from class priors.

### 4.2 General Remedy: Per-Class Style MMD

The remedy we propose is architecture-agnostic and applies to any factorized model with access to labels at training time. The standard, insufficient regularizer matches the marginal:

$$\mathcal{L}_{\text{style}} = \text{MMD}_{\text{RBF}}(\{z_s^i\}_{i=1}^b, \{\varepsilon_j\}_{j=1}^b), \quad \varepsilon_j \sim \mathcal{N}(0, I). \quad (2)$$

Our proposed fix instead matches each class conditional independently:

$$\mathcal{L}_{\text{style-cls}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_s^i : y_i=k\}, \{\varepsilon_j\}_{|y_i=k|}). \quad (3)$$

Where Eq. 2 has zero gradient incentive to reduce inter-class separation – the mixture can match  $\mathcal{N}(0, I)$  while every conditional remains class-structured (Proposition 1) – Eq. 3 penalizes exactly that failure mode, by construction. Nothing about this term is specific to hyperspherical semantic variables, ResNet encoders, or any other choice made in the case-study instantiation of Section 4.3: it requires only a labeled batch and a style latent to regularize.

**Scope of supervision.** Because  $\mathcal{L}_{\text{style-cls}}$  is computed per class, it requires labels at training time and is therefore a label-guided, supervised regularizer, not an unsupervised disentanglement method in the sense of  $\beta$ -TCVAE or FactorVAE. This is a different resource assumption than the four unsupervised baselines in Section 6.1, and we do not present the comparison between them as a like-for-like ablation of *how* to achieve independence, only as evidence that both unsupervised and supervised approaches leave measurable leakage. The *diagnostic* of Section 3.5, by contrast, requires only a labeled evaluation set and can be applied post hoc to any trained model regardless of how it was supervised, which is what makes it usable to audit the four unsupervised baselines in the first place.

### 4.3 Case-Study Instantiation: F-CS-WAE

To test the diagnostic and the remedy on a concrete model, we use F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder), which separates the latent into a hyperspherical semantic variable  $z_c \in \mathbb{S}^{d_c-1}$ , aligned to class-conditional Spherical Cauchy priors via a Möbius reparameterization [21], and a Euclidean style variable  $z_s \in \mathbb{R}^{d_s}$  regularized by Eqs. 2–3. A shared ResNet-18 backbone feeds two heads producing  $(\mu_c, \rho_c, \mu_s, \log \sigma_s^2)$ ;  $z_c$  and  $z_s$  are concatenated and decoded by a residual upsampling network. Full architectural detail is in Appendix A. The total training objective additionally includes a supervised semantic MMD toward the class prior ( $\mathcal{L}_{\text{class}}$ ), an aggregated semantic MMD ( $\mathcal{L}_{\text{agg}}$ ), and an auxiliary classification loss on  $\mu_c$  ( $\mathcal{L}_{\text{cls}}$ ) that prevents semantic collapse; these terms shape  $z_c$  for clustering and are not part of the style-leakage remedy itself. Coefficients are ramped over a four-phase, 300-epoch curriculum in which per-class style regularization begins in phase B, after a 50-epoch reconstruction-only phase A during which the decoder does see real  $(z_c, z_s)$  pairs with no style regularization; Experiment 7 (Section 6.8) tests whether removing this warmup materially changes the leakage, and finds it does not (full schedule in Appendix C).

### 4.4 Inference and Generation

**Clustering.** K-means on deterministic  $\mu_c$ , Hungarian-matched to compute ACC/NMI/ARI; labels are not used.

**Naive prior sampling.**  $z_c \sim p(z_c | y=k)$ ,  $z_s \sim \mathcal{N}(0, I)$ . Self-accuracy improves substantially when per-class style MMD is used (Section 6.4), though not to full reliability on its own.

**Class-conditional style prior (post-hoc).** Without per-class style MMD, estimate  $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$ ,  $\bar{\sigma}_s^{2(k)} = \text{Var}[\mu_s | y=k]$  from encoded training data, then sample  $z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)}))$  with  $\tau = 0.25$ . This is the second remedy we test, and it requires no retraining – only a forward pass over labeled training data to estimate  $\bar{\mu}_s^{(k)}$  and  $\bar{\sigma}_s^{2(k)}$ .

## 5 Case Study Setup

Before using F-CS-WAE to study leakage, we check that it is a reasonably competent model on its own terms – otherwise the leakage findings below would be unsurprising, since a degenerate model could leak trivially. We report this briefly, as a sanity check rather than a headline result.

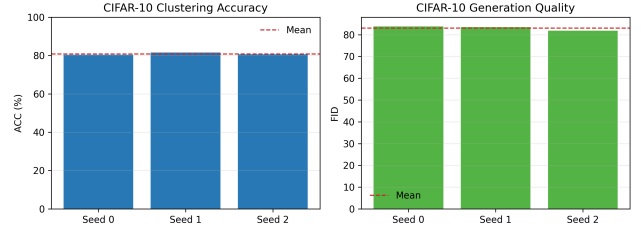
**Datasets.** CIFAR-10 [16] ( $3 \times 32 \times 32$ ,  $K=10$ ) is the primary clustering/generation benchmark. MNIST [18] and Fashion-MNIST [24] ( $1 \times 32 \times 32$ ,  $K=10$ ) are used for diagnostic analysis.

**Table 1:** CIFAR-10, single seed, against baselines matched to F-CS-WAE in backbone (ResNet-18) and total latent dimension ( $d=192$ ) [19]. Baselines are grouped by how the label is used: label-guided (label shapes the latent), conditional-generative (label conditions only the decoder), and unsupervised (no label). F-CS-WAE is 3-seed mean $\pm$ std.

| Method                                                   | ACC          | NMI          | FID         |
|----------------------------------------------------------|--------------|--------------|-------------|
| <b>F-CS-WAE (3-seed)</b>                                 | <b>0.809</b> | <b>0.650</b> | <b>83.0</b> |
| $\pm$ std                                                | .005         | .008         | .9          |
| <i>Label-guided (label shapes the latent)</i>            |              |              |             |
| AEWithCE                                                 | 0.665        | 0.611        | 144.3       |
| AEWithSupCon                                             | 0.541        | 0.599        | 275.5       |
| AEWithTriplet                                            | 0.506        | 0.502        | 281.4       |
| AEWithCenterLoss                                         | 0.243        | 0.122        | 235.8       |
| <i>Conditional-generative (label conditions decoder)</i> |              |              |             |
| ConditionalVAE                                           | 0.133        | 0.024        | 396.4       |
| ConditionalWAE-MMD                                       | 0.102        | 0.008        | 170.3       |
| GaussianClassPriorWAE                                    | 0.103        | 0.008        | 174.7       |
| <i>Unsupervised (no label)</i>                           |              |              |             |
| ResNetAE                                                 | 0.220        | 0.102        | 121.0       |

Table 1 compares F-CS-WAE against seven baselines matched in backbone and latent dimension, and receiving the same label access (except the unsupervised reference). The fair clustering comparison is against the label-guided autoencoders, which – like F-CS-WAE – push the label signal into the latent: F-CS-WAE (80.9%) exceeds the best of them (AEWithCE, 66.5%), and also has the lowest FID of any model in the table (83.0). The conditional-generative baselines cluster near chance (10–13%) because their label enters only at the decoder, leaving the encoder latent unshaped for clustering; we include them as the closest generative analogues, not as a clustering comparison. We read this as evidence that F-CS-WAE is competitive under matched supervision and not degenerate, but not as a controlled superiority claim: it is single-seed for the baselines, F-CS-WAE uses *both* latent-shaping and a generative prior where each baseline uses only one, and we did not equalize per-method hyperparameter tuning or compute (Section 7.1). On MNIST, the single-latent CS-WAE variant (one Spherical Cauchy semantic latent, no style variable) reaches ACC 0.8735, NMI 0.8863, ARI 0.8411, FID 26.95 – included for context in Table 3 below, since it has no style latent and so cannot leak by definition. On Fashion-MNIST, F-CS-WAE (single seed) reaches ACC 0.9341, NMI 0.8672, ARI 0.8633, FID 72.37, sitting between MNIST and CIFAR-10 – a pattern we return to in Section 6.5.

**Metrics used throughout.** Clustering: ACC, NMI, ARI. Reconstruction: SSIM, PSNR, LPIPS [26]. Sample quality: FID [7] (10,000 generated vs. 10,000 real). Generation self-accuracy: fraction of generated images the model’s trained auxiliary classifier assigns to the sampled class (a same-model evaluator whose limitations we discuss in Sec-



**Figure 1:** CIFAR-10 clustering and FID across three F-CS-WAE seeds. Variance is low.

tion 7.1). Leakage: global MMD,  $\Delta_{\text{inter}}$ , LP (HSIC defined but not reported, Section 3.5).

## 6 Experiments

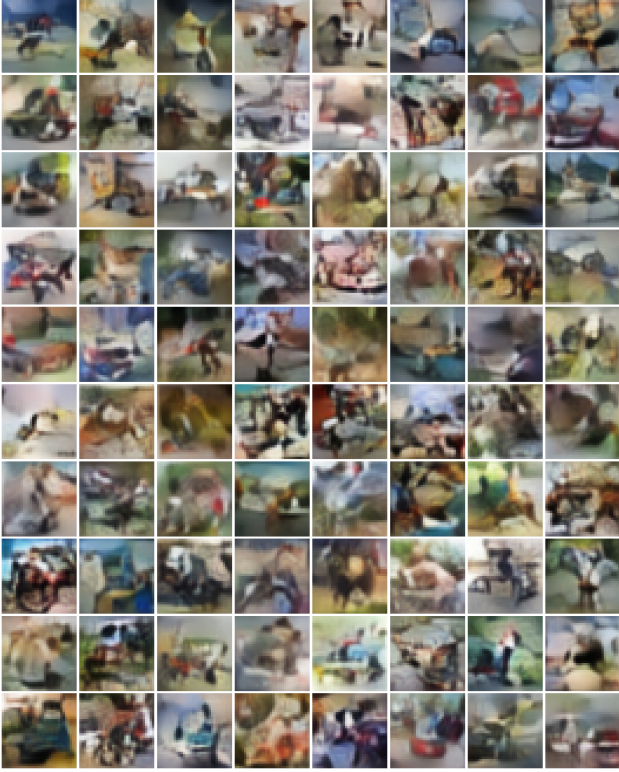
We organize the experiments by claim rather than by dataset: each subsection states what it tests, on which checkpoint, and what would falsify it.

### 6.1 Experiment 1: Marginal Metrics Hide Leakage

*Claim tested:* near-zero global MMD does not imply  $z_s \perp y$ , across model families, not just for F-CS-WAE.

We apply the diagnostic of Section 3.5 to four unsupervised baselines (VAE, WAE-MMD,  $\beta$ -TCVAE, FactorVAE) and F-CS-WAE on MNIST. The four baselines are purpose-built for this diagnostic only, with a lightweight 3-layer convolutional encoder trained for 100 epochs, and are separate from the matched-supervision baselines used for the CIFAR-10 clustering comparison in Section 5, which follow a different protocol (Appendix B). F-CS-WAE, by contrast, uses its full ResNet-18 encoder trained for 300 epochs (Section 4.3), so architecture and training budget are not matched across the five rows of Table 2 and the  $\Delta_{\text{inter}}$  ranking between F-CS-WAE and the baselines should not be read as a controlled comparison. What the mismatch does not undermine is the *within-model* evidence: for every row individually, global MMD is decoupled from  $\Delta_{\text{inter}}$  and LP regardless of that row’s encoder capacity or training length, which is the comparison the diagnostic actually depends on.

*Result.* All four baselines show LP far above chance (74–93%) despite near-zero global MMD. WAE-MMD has the largest global MMD among baselines (0.0576, 30–300 $\times$  the others), suggesting its own regularizer had not converged, and correspondingly the largest baseline  $\Delta_{\text{inter}}$  (9.38) and LP (93%).  $\beta$ -TCVAE and FactorVAE, whose training objectives explicitly penalize latent dependence, show the smallest  $\Delta_{\text{inter}}$  among the five models (2.30, 2.27) – but LP stays far above chance (74–76%): penalizing total correlation between latent *dimensions* reduces, but does not eliminate, dependence between the latent and the *label*, which is a different quantity. F-CS-WAE without the proposed remedy shows the most extreme leakage of the five ( $\Delta_{\text{inter}} = 6.27$ ,



**Figure 2:** CIFAR-10 class-conditional prior samples (one row per class). FID 83 places these in the recognizable-but-blurry regime typical of L1/LPIPS-trained decoders without an adversarial loss, well short of photorealistic (Section 7).

LP= 100%), which Experiment 2 shows corresponds to a concrete generation failure, not just an elevated statistic.

Appendix D visualizes  $z_s$  for all six rows with t-SNE: the leakage is visually obvious only for WAE-MMD and F-CS-WAE (the two largest  $\Delta_{\text{inter}}$  rows), while for VAE,  $\beta$ -TCVAE, and FactorVAE it is statistically real (LP 74–87%) but not visually apparent in a 2D projection – evidence that a plot is not a substitute for the quantitative diagnostic.

## 6.2 Experiment 2: Leakage Breaks Class-Conditional Generation

*Claim tested:* on a concrete checkpoint, leakage is not just a statistic – it corresponds to generation the auxiliary classifier judges as wrong.

*Result.* F-CS-WAE reaches 99.15% clustering accuracy and SSIM 0.983, yet class-conditional generation self-accuracy under naive sampling is only 15%: near-perfect scores on every standard metric coexist with a decoder that, when handed the cross-class  $(z_c, z_s)$  pairs that naive sampling produces, does not generate the requested class.

**Table 2:** Experiment 1. Conditional style leakage across five model families on MNIST. Despite near-zero global MMD in every case, inter-class style separation  $\Delta_{\text{inter}}$  is large and linear probing (LP) recovers class information well above the 10% chance rate. The last row adds the remedy of Experiment 3. Self-ACC uses naive  $z_s \sim \mathcal{N}(0, I)$  sampling.

| Model              | MMD ↓  | $\Delta_{\text{inter}}$ ↓ | LP ↓   | Self-ACC ↑ |
|--------------------|--------|---------------------------|--------|------------|
| VAE                | 0.0002 | 2.86                      | 87.4%  | N/A        |
| WAE-MMD            | 0.0576 | 9.38                      | 93.0%  | N/A        |
| $\beta$ -TCVAE     | 0.0002 | 2.30                      | 73.9%  | N/A        |
| FactorVAE          | 0.0009 | 2.27                      | 75.9%  | N/A        |
| F-CS-WAE           | 0.0013 | 6.27                      | 100.0% | 0.15       |
| F-CS-WAE+per-class | 0.0015 | 1.46                      | 45.6%  | 0.70       |

**Table 3:** Experiment 2. MNIST case study, single seed (seed 0). F-CS-WAE vs. the single-latent CS-WAE, which has no style latent to leak.

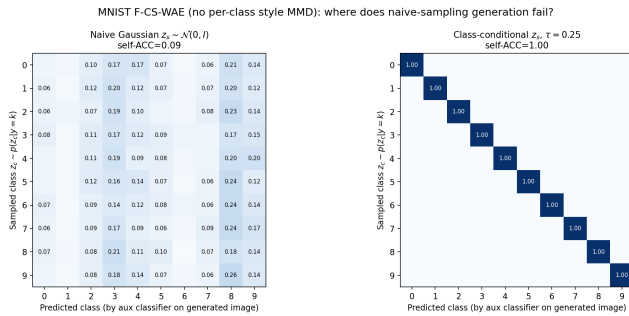
| Method   | ACC    | NMI    | ARI    | FID   | $\Delta_{\text{inter}}$ |
|----------|--------|--------|--------|-------|-------------------------|
| CS-WAE   | 0.8735 | 0.8863 | 0.8411 | 26.95 | –                       |
| F-CS-WAE | 0.9915 | 0.9758 | 0.9813 | 73.99 | 6.27                    |

Figure 3 shows the failure is systematic rather than random: naive sampling does not spread errors uniformly across the nine wrong classes but collapses toward a small set of “attractor” digits (notably 8) regardless of which class was requested. Figure 4 makes the same point at the pixel level: naive-prior samples are legible digits, just frequently the *wrong* digit, which is consistent with the decoder resolving an out-of-training-distribution  $(z_c, z_s)$  pair by falling back on whichever class its style input most resembles.

## 6.3 Direct Evidence: Decoder Sensitivity to Leaked Style

*Claim tested:* the linear probe of Section 3.5 shows  $z_s$  contains recoverable class information; it does not show the decoder *uses* that information to decide what to generate. We test this directly with two additional measurements on the same  $\delta=0$  and  $\delta=1$  F-CS-WAE checkpoints as Table 2.

**The full conditional structure, not just its mean.**  $\Delta_{\text{inter}}$  summarizes class separation as a single mean-to-mean distance; it says nothing about how far each class conditional  $q(z_s | y=k)$  itself sits from the prior it is supposed to match. We compute  $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$  per class and pairwise  $\text{MMD}^2(q(z_s | y=i), q(z_s | y=j))$  for all class pairs, on MNIST and CIFAR-10, at  $\delta=0$  and  $\delta=1$ . At  $\delta=0$ , mean per-class MMD is  $21\times$  the global MMD on MNIST (0.0186 vs. 0.00089) and  $15\times$  on CIFAR-10 (0.0085 vs. 0.00056) – Figure 5 makes the point visually: every one of the ten per-class bars sits far above the global-MMD reference line the marginal regularizer actually optimizes. At  $\delta=1$  the gap nar-

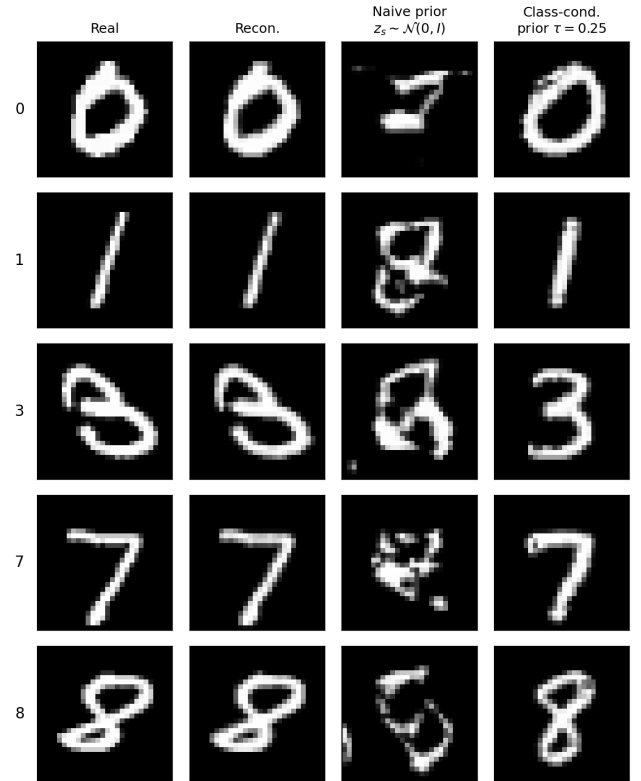


**Figure 3:** Confusion between the sampled class  $y=k$  and the class the auxiliary classifier assigns to the generated image. Naive Gaussian sampling (left) does not fail randomly—probability mass collapses onto a handful of “attractor” digits (notably 8) regardless of the intended class. The class-conditional prior (right,  $\tau=0.25$ ) is a clean diagonal.

rows but does not close (MNIST  $4.4\times$ ; CIFAR-10  $5.3\times$ ), the same “reduces but does not eliminate” pattern Experiment 3 reports for  $\Delta_{\text{inter}}$  and LP, now shown to hold for the full per-class MMD, not only its mean-separation proxy.

**Latent swap: a controlled intervention.** For every ordered class pair  $(a, b)$  we take  $\mu_c$  from a real test image of class  $a$  and  $\mu_s$  from an independently drawn real test image of class  $b$ , decode  $\text{Dec}(\mu_c, \mu_s)$ , re-encode the result, and classify it with the same auxiliary classifier used for self-accuracy elsewhere. This is a stronger test than naive-prior sampling: it uses only real, individually-encoded outputs rather than prior samples, so a failure to recover class  $a$  cannot be attributed to an out-of-support prior draw – though the specific  $(z_c, z_s)$  pairing itself may lie outside the joint support the decoder saw during training, since  $z_c$  and  $z_s$  here come from two different real images. What it isolates is what the decoder does with a  $(z_c, z_s)$  pair it never saw paired together, using components that are each individually real. We report the diagonal-excluded means of  $P(\text{pred}=a)$  (content-following) and  $P(\text{pred}=b)$  (style-following).

**Result.** On MNIST at  $\delta=0$ , off-diagonal content-following is 0.1% and style-following is 99.2% (Figure 6, top): the decoder does not merely have access to leaked class information in  $z_s$ , it relies on that information almost exclusively to decide what to draw, discarding  $z_c$  almost entirely off-diagonal. Figure 7 shows why: every row (fixed  $z_c$ ) produces visually identical output regardless of which class donated  $z_c$ ; only the column (the  $z_s$  donor) determines the digit. At  $\delta=1$ , content-following rises to 60.9% and style-following falls to 18.8% – a genuine change in what the decoder conditions on under a controlled intervention, not merely a reduction in a correlational statistic. On CIFAR-10 the same qualitative flip occurs but is weaker (content-following  $4.2\% \rightarrow 20.0\%$ , style-following  $68.8\% \rightarrow 38.5\%$ ; heatmaps and grid in Appendix D, Figures 15–16), consistent with the cross-dataset asymmetry of Experiment 4 and Experiment 5: CIFAR-10’s



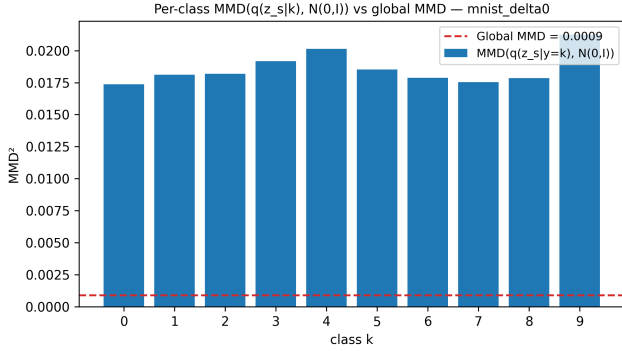
**Figure 4:** Real image, posterior reconstruction, naive-prior sample, and class-conditional-prior sample, for five representative digits. Reconstructions are visually indistinguishable from the real image (SSIM 0.983); naive-prior samples are legible digits, but frequently the *wrong* digit (e.g. row “0”  $\rightarrow$  looks like a “3”; row “1”  $\rightarrow$  looks like a “9”); the class-conditional prior restores the intended identity.

higher intra-class diversity means the decoder was never as exclusively dependent on  $z_s$  for identity to begin with, so there is proportionally less for the remedy to undo. The pairwise conditional-MMD structure behind Figure 5’s per-class bars, for both datasets, is likewise in Appendix D (Figure 14).

## 6.4 Experiment 3: Per-Class MMD Reduces but Does Not Eliminate Leakage

**Claim tested:** the remedy of Section 4.2 works in the direction intended, but does not drive leakage to zero, and reaching 100% self-accuracy requires combining it with the post-hoc prior.

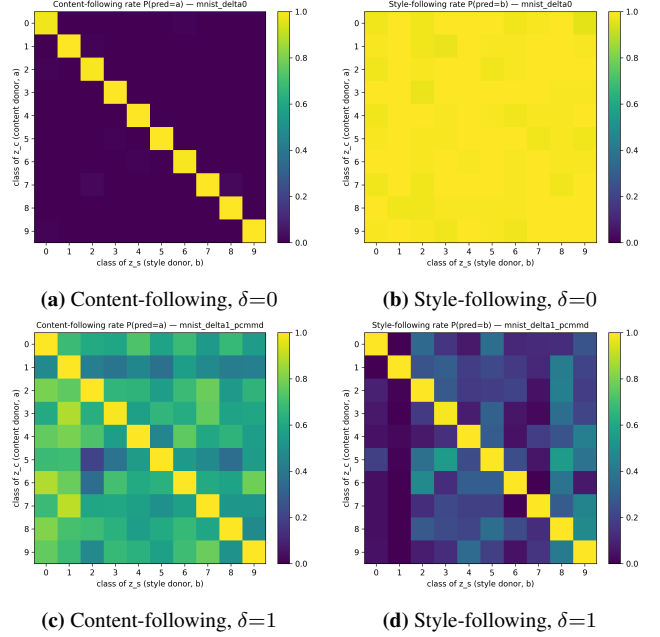
**Result.** From Table 2, per-class style MMD cuts F-CS-WAE’s  $\Delta_{\text{inter}}$  by 77% ( $6.27 \rightarrow 1.46$ ) and LP by 54 points ( $100\% \rightarrow 45.6\%$ ) – still far above the 10% chance rate, and higher than three of the four unsupervised baselines in the same table. Naive-sampling self-accuracy rises from 0.15 to 0.70, not 1.00: the remedy reduces how much class information  $z_s$  carries, but a linear probe can still recover the label well above chance, and naive sampling still fails on roughly 3 in 10 generations. Theorem 1 gives a structural



**Figure 5:** MNIST,  $\delta=0$ : per-class  $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$  (bars) against the global MMD the marginal regularizer optimizes (dashed line). Every class sits far above the line the model was actually trained to minimize.

reason this remedy is necessarily partial:  $\mathcal{L}_{\text{style-cl}}^{\text{cls}}$  (Eq. 3) drives terms (1)–(2) toward zero by construction, but places no direct gradient pressure on term (4),  $I_q(z_c; z_s | y)$  – the within-class dependence the decoder can still exploit even once every  $q(z_s | y=k)$  individually matches  $\mathcal{N}(0, I)$ . The JointMMD proxy for term (4) (Section 3.5) is consistent with this: it shows no observable reduction from  $\delta=0$  to  $\delta=1$  (0.0037  $\rightarrow$  0.0043 on MNIST) while  $\Delta_{\text{inter}}$  and LP fall sharply – as expected if the remedy leaves term (4) unaddressed, though (per the calibration caveats in Section 3.5) this single-seed proxy cannot by itself establish that the underlying dependence is unchanged. Reaching 100% requires *combining* both remedies: applying the post-hoc class-conditional prior ( $\tau=0.25$ , Section 6.6) on top of the  $\delta=1$  checkpoint, rather than relying on either alone, which we verified directly on this checkpoint (naive 0.70  $\rightarrow$  class-conditional 1.00, same protocol as Table 4). We read the two remedies as complementary rather than redundant: per-class style MMD reduces how much class information  $z_s$  carries, and the post-hoc prior corrects for what remains.

**Why does combining remedies still work at  $\delta=1$ ?** The class-conditional sampling strategies above are themselves fit to per-class statistics of  $z_s$ , so it is natural to ask whether they remain distinguishable once  $\delta=1$  has pulled every  $q(z_s | y=k)$  toward  $\mathcal{N}(0, I)$ . We test this in latent space alone, with no decoding: an independent linear probe trained on real  $z_s$  (held-out accuracy 98.7% at  $\delta=0$ , only 32.3% at  $\delta=1$ , consistent with the LP drop above) is asked to classify synthetic  $z_s$  draws from each sampling strategy. At  $\delta=0$  every class-conditional strategy is  $\geq 99.8\%$  self-consistent by this measure. At  $\delta=1$ , consistency collapses: the class-mean strategy falls to 80.0% and the genuinely-distributional strategies ( $\tau=0.25$ ,  $\tau=0.50$ , empirical bank) fall to 15–28% – lower than at  $\delta=0$ , even though decoder self-accuracy under the same strategies reaches 100% (verified above). The resolution is Section 6.3’s controlled latent-swap intervention: at



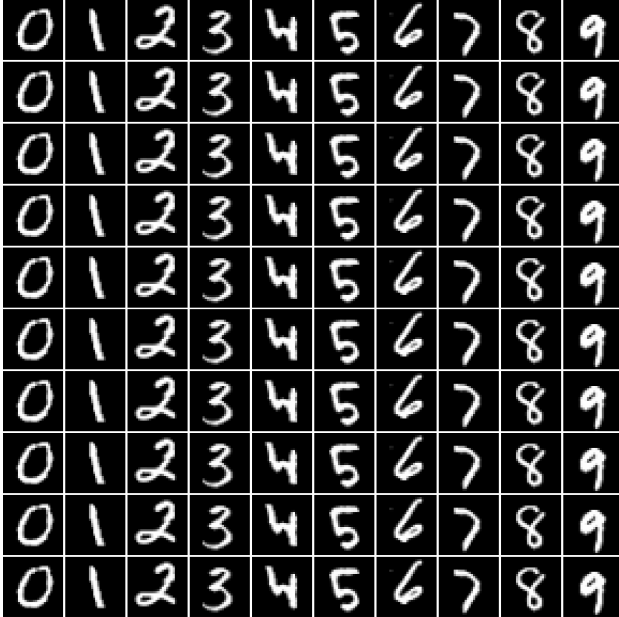
**Figure 6:** MNIST latent-swap intervention: rows are the class donating  $z_c$  (content), columns the class donating  $z_s$  (style). At  $\delta=0$  (top), off-diagonal output is almost never the content class (mean 0.1%) and almost always the style class (mean 99.2%): the decoder has learned to read identity off  $z_s$ . At  $\delta=1$  (bottom), this flips substantially (content 60.9%, style 18.8%).

$\delta=1$  the decoder has shifted toward reading identity off  $z_c$  rather than  $z_s$  (content-following 0.1%  $\rightarrow$  60.9%), so the specific  $z_s$  value supplied at generation time matters less than it did at  $\delta=0$ . Combining remedies reaches 100% self-accuracy not because the sampled  $z_s$  becomes more class-distinguishable in latent space – by this measure it becomes *less* distinguishable – but because the decoder itself has become less sensitive to which  $z_s$  it receives.

## 6.5 Experiment 4: Leakage Follows the Assumed Intra-Class Diversity Ordering, Self-Accuracy Does Not

*Claim tested:* Section 3.4 predicts leakage severity should decrease as intra-class visual diversity increases, under the *assumed* ordering MNIST (least diverse)  $>$  Fashion-MNIST  $>$  CIFAR-10 (most diverse). We take this ordering as a qualitative prior – we do not quantify intra-class diversity with an independent metric (e.g. within-class feature covariance or intrinsic dimensionality), so this experiment tests consistency with an assumed ordering, not a measured correlation. We evaluate on the  $\delta=0$  F-CS-WAE checkpoint for all three datasets, using the same diagnostic throughout.

*Result.*  $\Delta_{\text{inter}}$  and LP are monotone in the assumed ordering: MNIST (6.27, 100.0%)  $>$  Fashion-MNIST (4.86, 97.7%)  $>$  CIFAR-10 (4.19, 89.9%) – consistent with the diversity hypothesis, though with only three datasets and no

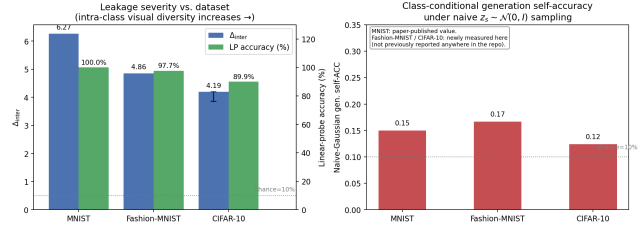


**Figure 7:** MNIST latent-swap grid,  $\delta=0$ : row =  $z_c$  donor class, column =  $z_s$  donor class. Every row is visually identical – identity is set entirely by the column.

independent diversity measurement we read this as suggestive rather than an established correlation. Naive-Gaussian generation self-accuracy does not: MNIST 0.15, Fashion-MNIST 0.17, CIFAR-10 0.12 – CIFAR-10 is lowest, not highest. The two families of metrics differ in what they are computed on, and that difference explains the disagreement rather than undermining either measurement.  $\Delta_{\text{inter}}$  and LP operate directly on  $z_s$  and therefore isolate the dependency we care about. Self-accuracy is measured on the generated *image*, mediated by an auxiliary classifier, so it additionally inherits how confusable the real classes already are under blurry generation – a dataset-difficulty effect with nothing to do with conditional style leakage, and one that CIFAR-10 has on top of leakage. We see no principled reason to discard the metric that disagrees, and doing so would let us report a cleaner story than the data supports; the disagreement instead tells us that generation self-accuracy, on its own, is not a reliable cross-dataset proxy for leakage severity, even though it remains the right metric for asking whether a specific checkpoint’s generation is repaired (Section 6.6).

## 6.6 Experiment 5: Repair Does Not Transfer Uniformly Across Datasets

*Claim tested:* the class-conditional sampling recipe validated on MNIST (Experiment 2) transfers to CIFAR-10 with the same effectiveness.



**Figure 8:** Left:  $\Delta_{\text{inter}}$  and linear-probe accuracy across datasets, ordered by increasing intra-class visual diversity—both decrease monotonically, matching the hypothesis. Right: naive-Gaussian generation self-accuracy does *not* follow the same order (CIFAR-10 is lowest, not highest); see text.

**Table 4:** Experiment 5, MNIST. Generation under five  $z_s$  sampling strategies on the  $\delta=0$  checkpoint;  $z_c$  always from the class prior.

| Style sampling $z_s$     | Self-ACC     | Conf.        | Div.         |
|--------------------------|--------------|--------------|--------------|
| Global Gaussian          | 0.150        | 0.934        | 0.151        |
| Class mean               | 1.000        | 1.000        | 0.132        |
| Class diag., $\tau=0.50$ | 1.000        | 0.990        | 0.139        |
| Class diag., $\tau=0.25$ | <b>1.000</b> | <b>1.000</b> | 0.135        |
| Empirical style bank     | 1.000        | 1.000        | <b>0.182</b> |

On MNIST, every class-conditional strategy restores perfect self-accuracy (Table 4); the class-conditional diagonal Gaussian with  $\tau=0.25$  is a true parametric sampler with perfect class consistency and better diversity than the class-mean point estimate. One confound deserves emphasis: the class-conditional strategies differ from the global  $\mathcal{N}(0, I)$  baseline in *two* ways at once – a per-class mean shift *and* a variance shrinkage ( $\tau=0.25$ , i.e. sampling much closer to the per-class posterior mean). Part of the self-accuracy gain may therefore come from drawing  $z_s$  nearer the posterior manifold (where the decoder is well-behaved) rather than from class conditioning per se. We do not fully separate these here; the controls that would (a  $\tau=0.25$  global fitted Gaussian, a per-class mean with global covariance, and a  $\tau=1$  class-conditional Gaussian) are the natural next step, and we note that the class-mean point estimate – pure conditioning with near-zero variance – already reaches 100%, so conditioning is clearly sufficient, even if its separation from variance shrinkage is not isolated.

*Result.* The  $\tau=0.25$  recipe that is perfect on MNIST reaches only 68.1% on CIFAR-10 (Table 5), and this is not a sample-size artifact: the qualitative grid (Figure 9) shows why the number alone overstates the fix – samples within a class collapse toward a single low-diversity image, and several classes are not clearly identifiable by eye even where the classifier scores them correctly. Only the empirical style bank, which resamples real encoded style codes rather than fitting a per-class Gaussian, approaches MNIST-level self-accuracy (95.9%), suggesting CIFAR-10’s per-class style distribution is not well approximated by a unimodal Gaussian,

**Table 5:** Experiment 5, CIFAR-10. Same five sampling strategies,  $\delta=0$  checkpoint. Unlike MNIST, the parametric class-conditional strategies plateau well below 100%.

| Style sampling $z_s$     | Self-ACC     | Conf. | Div.         |
|--------------------------|--------------|-------|--------------|
| Global Gaussian          | 0.119        | 0.911 | 0.187        |
| Class mean               | 0.700        | 0.996 | 0.079        |
| Class diag., $\tau=0.50$ | 0.687        | 0.970 | 0.138        |
| Class diag., $\tau=0.25$ | 0.681        | 0.975 | 0.101        |
| Empirical style bank     | <b>0.959</b> | 0.988 | <b>0.241</b> |

unlike MNIST’s. The direct evidence favors a distributional rather than a dependence-theoretic explanation: the empirical style bank – which differs from the Gaussian samplers only in matching the *shape* of  $q(z_s | y=k)$ , not in supplying any  $z_c$ -conditional value – recovers almost all of the lost self-accuracy, which is what one expects if CIFAR-10’s per-class  $q(z_s | y=k)$  is simply multimodal enough that a unimodal Gaussian undershoots it. We had also considered a term-4 explanation (CIFAR-10’s greater intra-class diversity producing larger within-class  $z_c$ - $z_s$  dependence, which no  $z_c$ -independent sampler could capture), but we cannot adjudicate it with the current JointMMD proxy: its MNIST and CIFAR-10 values are close (0.0037 vs. 0.0036 at  $\delta=0$ ), yet, as noted in Section 3.5, uncalibrated MMD magnitudes on two different latent distributions are not directly comparable, so this neither supports nor rules out the term-4 account. We therefore report the distributional reading as the one the evidence actually supports and leave the term-4 question to a calibrated future measurement.

We treat this as a genuine limitation of the proposed remedy rather than omit it: the diagnostic of Experiment 1 and Experiment 4 generalizes across datasets more cleanly than the fix does, and a reader should not extrapolate the MNIST self-accuracy numbers to other datasets without checking.

## 6.7 Experiment 6: Dataset-Dependent Clustering-Leakage Trade-off

*Claim tested:* per-class style MMD’s cost to clustering accuracy is not a fixed price of the fix, but depends on how much class information the dataset had pushed into  $z_s$  in the first place.

*Result.* The original two-point comparison remains accurate as far as it goes: on CIFAR-10, per-class style MMD is close to free at  $\delta=1$  (81.49%  $\rightarrow$  80.00% ACC; the 3-seed mean without the regularizer is  $80.87 \pm 0.52\%$ , Table 1),  $\Delta_{\text{inter}}$  drops 67%, and naive-sampling FID improves (83.59  $\rightarrow$  74.42); on MNIST the leakage reduction is larger in relative terms (77%) but clustering accuracy drops substantially (99.15%  $\rightarrow$  85.92%).

The four additional  $\delta$  values in Table 6 show this two-point picture, while accurate, is incomplete.  $\Delta_{\text{inter}}$  and naive FID both decrease close to monotonically with  $\delta$  on both

**Table 6:** Effect of per-class style MMD at six  $\delta$  values, both datasets, single seed per row.  $\delta=0$  and  $\delta=1$  rows reproduce Tables 2/3; the four intermediate values are new.

| Dataset ( $\delta$ )       | ACC    | $\Delta_{\text{inter}}$ | Naive FID |
|----------------------------|--------|-------------------------|-----------|
| MNIST ( $\delta=0$ )       | 0.9915 | 6.27                    | 73.99     |
| MNIST ( $\delta=0.03$ )    | 0.9944 | 3.27                    | 62.80     |
| MNIST ( $\delta=0.1$ )     | 0.9898 | 2.12                    | 52.61     |
| MNIST ( $\delta=0.3$ )     | 0.9936 | 1.55                    | 59.44     |
| MNIST ( $\delta=1$ )       | 0.8592 | 1.46                    | 43.36     |
| MNIST ( $\delta=3$ )       | 0.9929 | 1.18                    | 41.54     |
| CIFAR-10 ( $\delta=0$ )    | 0.8149 | 4.19                    | 83.59     |
| CIFAR-10 ( $\delta=0.03$ ) | 0.8111 | 2.48                    | 79.11     |
| CIFAR-10 ( $\delta=0.1$ )  | 0.8100 | 2.03                    | 77.50     |
| CIFAR-10 ( $\delta=0.3$ )  | 0.7148 | 1.59                    | 72.05     |
| CIFAR-10 ( $\delta=1$ )    | 0.8000 | 1.28                    | 74.42     |
| CIFAR-10 ( $\delta=3$ )    | 0.7994 | 0.95                    | 76.11     |

datasets – the leakage-reduction mechanism behaves as expected across the whole range, not just at the two endpoints we originally measured. Clustering ACC does not move with it the same way: on MNIST,  $\delta \in \{0.03, 0.1, 0.3, 3\}$  all sit at 99–99.4%, statistically indistinguishable from  $\delta=0$ , and  $\delta=1$  is the *only* point with a large drop; on CIFAR-10,  $\delta \in \{0, 0.03, 0.1, 1, 3\}$  all sit at 80–81%, and  $\delta=0.3$  is the only point with a large drop (71.5%). We ran one seed per  $\delta$ , so we cannot separate a genuine non-monotonic effect of  $\delta$  on clustering from this particular run’s initialization at that one point; resolving that would require repeating each  $\delta$  across multiple seeds, which is beyond this draft’s compute budget (Section 7.1). What the sweep does establish, which the two-point comparison could not, is directional: leakage reduction is a smooth function of  $\delta$  on this evidence, while clustering accuracy is not, and neither dataset shows a clean monotonic trade-off curve between the two at a single seed per point. The original interpretation – that the endpoint cost tracks how much a dataset’s model depended on the leak – remains consistent with the  $\delta=0$ -vs- $\delta=1$  comparison it was based on, but should not be read as describing a smooth relationship across  $\delta$ , which this sweep shows does not hold.

## 6.8 Experiment 7: Leakage Survives Capacity, Curriculum, Prior-Geometry, and Supervision Perturbations

*Claim tested:* the extreme leakage found at  $\delta=0$  (Section 6.2:  $\Delta_{\text{inter}}=6.26$ , LP= 100%) is a genuine property of the factorized-sampling failure this paper studies, not an artifact of specific design choices in F-CS-WAE – in particular the style latent being larger than the semantic latent ( $d_s=128 > d_c=64$ ), the 50-epoch reconstruction-only warmup before any style regularization, the choice of semantic-prior geometry (Spherical Cauchy), or the auxiliary classifier. Each is independently plausible as the real explanation for why  $z_s$  absorbs so much

**Table 7:** Experiment 7. Single-variable perturbations from the  $\delta=0$  checkpoint, seed 0, each removing one candidate confound, on both datasets. Same diagnostic protocol as Table 2 (`compute_leakage_diagnostics.py`,  $n=2048$ , seed 0).

| Perturbation               | MNIST                   |               | CIFAR-10                |              |
|----------------------------|-------------------------|---------------|-------------------------|--------------|
|                            | $\Delta_{\text{inter}}$ | LP            | $\Delta_{\text{inter}}$ | LP           |
| <b>None (baseline)</b>     | <b>6.26</b>             | <b>100.0%</b> | <b>3.80</b>             | <b>86.9%</b> |
| Balanced dims              | 4.79                    | 99.9%         | 2.11                    | 52.4%        |
| Smaller style ( $d_s=32$ ) | 3.28                    | 92.6%         | 1.68                    | 44.0%        |
| No warmup                  | 5.53                    | 100.0%        | 3.78                    | 88.3%        |
| Gaussian prior             | 5.35                    | 99.6%         | 3.82                    | 74.5%        |
| vMF prior                  | 5.69                    | 100.0%        | 3.03                    | 83.0%        |
| No classifier              | 5.09                    | 97.1%         | 3.15                    | 67.1%        |

class information on this checkpoint specifically, and each could plausibly matter differently depending on how class-structured a dataset’s images already are (Section 6.5), so we run all six on *both* MNIST and CIFAR-10.

**Result.** LP never drops below 92.6% on MNIST or 44.0% on CIFAR-10 across six independent perturbations; neither ever approaches chance (10%), and CIFAR-10’s minimum (44.0%) remains above the level per-class style MMD reaches at  $\delta=1$  on the same dataset (35.8%, measured on the same checkpoint used for the term-4 JointMMD values in Section 3.5). On MNIST every perturbation is mild (LP falls by at most 7.4 points); on CIFAR-10 the same six perturbations produce a much wider spread (1.4-point *increase* to a 42.9-point drop, Figure 11). The ranking by effect size is consistent across both datasets – smaller style capacity is the single largest effect on MNIST (7.4 points) and on CIFAR-10 (42.9 points), followed by balanced dims (0.1 and 34.5 points) – but the two capacity perturbations are not uniquely special on CIFAR-10: removing the auxiliary classifier (19.8 points) and the Gaussian prior (12.4 points) also produce sizable drops there, while no-warmup (−1.4, i.e. no effect) and the vMF prior (3.9) barely move it. Removing the auxiliary classifier also collapses MNIST clustering ACC from 99% to 55.4% on this checkpoint (confirming its documented anti-collapse role) without removing the leakage it was never designed to address. Two readings follow. First, capacity is a real, dataset-dependent contributing factor – consistent with Section 3.4’s prediction that severity tracks how much class-specific structure a dataset forces into  $z_s$  – and a paper that checked only MNIST would have significantly understated how much a single hyperparameter can move this number. We also flag a confound we do not disentangle: the capacity perturbations that reduce LP most also reduce the model’s representational capacity, and removing the classifier collapses clustering, so part of each LP drop may reflect a weaker or degenerate representation rather than genuine style-invariance; we report LP and  $\Delta_{\text{inter}}$  only, not matched reconstruction/FID/ACC per row, so we cannot separate these. Second, even so, no single tested perturbation drives LP anywhere near chance – CIFAR-10’s minimum

(44.0%) is still more than four times the 10% floor – so no one design choice we tested *eliminates* the leakage of Section 6.2 or Section 5, even though capacity clearly modulates its magnitude. This is single-seed evidence (Section 7.1); we have not tested perturbations jointly (e.g. smaller style *and* no classifier together) or repeated any point across seeds.

## 7 Discussion

**Why CIFAR-10 is milder than MNIST, with a caveat.** CIFAR-10’s higher intra-class visual diversity weakens conditional style dependence, and  $\Delta_{\text{inter}}$ /LP confirm this cleanly (Experiment 4). But Experiment 5 shows the practical remedy does not inherit this cleanly: the class-conditional Gaussian prior that perfectly fixes MNIST generation only partially fixes CIFAR-10 generation, and CIFAR-10 samples remain visibly blurry (FID  $\sim 83$ –84, well above photorealistic ranges). We avoid describing CIFAR-10 naive-sampling output as “visually plausible”; at this FID the more accurate description is “recognizable but not sharp,” and under the  $\tau=0.25$  class-conditional prior several classes are not reliably identifiable by eye despite scoring well on the auxiliary classifier (Figure 9).

**Global vs. conditional regularization.** Global aggregate matching is standard in WAEs and disentanglement models; our results show it is insufficient whenever the style variable must be class-invariant for generation. Per-class style MMD is a supervised analogue of total correlation penalties, targeting the single dependency  $z_s \not\perp y$  directly rather than dependence among latent dimensions in general.

**The reconstruction–generation gap.** The MNIST checkpoint reaches SSIM 0.983 yet FID 73.99 under naive prior sampling (Figure 4). Reconstruction quality measures fidelity on the posterior manifold, not the prior; evaluation of generative clustering models should report both, since Section 6.1 shows the two can diverge substantially.

### 7.1 Limitations

We list what this paper does *not* establish, rather than leave it implicit; several of these are previewed in the Scope paragraph of Section 1 and restated here in full. We consider the first two the most consequential. **(0) Same-model evaluation.** Both generation self-accuracy and the latent-swap content/style-following rates are computed with the model’s own jointly-trained auxiliary classifier, and the swap additionally re-encodes generated images through the same encoder; a bias shared between the encoder, classifier, and decoder could therefore inflate these numbers in a correlated way. We do not use an independently-trained external classifier, agreement between two such classifiers, direct pixel-space classification without re-encoding, or human annotation, any

of which would make the self-accuracy and swap results independent of the model under test. This is the single largest gap in the evaluation, and the swap’s 99.2% style-following on MNIST should be read as a strong within-model result pending external confirmation. **(0’) JointMMD calibration.** The term-4 proxy (Section 3.5) is a single biased estimate per checkpoint with no permutation null, bandwidth-sensitivity analysis, per-block normalization, or seed replication, so it can support only the weak, hedged claims we draw from it. (1) Baseline comparisons (Table 1) are single-seed for the baselines; F-CS-WAE uses both latent-shaping and a generative prior where each of the seven matched-supervision baselines uses only one, and we did not equalize per-method tuning or compute, so the CIFAR-10 comparison is evidence of competitiveness and non-degeneracy, not a controlled superiority claim. (2) The clustering-leakage trade-off (Table 6, Figure 10) now spans six  $\delta$  values per dataset rather than two, but each is a single training run; the non-monotonic ACC pattern we observe (Section 6.7) may reflect true  $\delta$ -dependent instability at specific values or ordinary seed variance we cannot distinguish without repeating each  $\delta$  across multiple seeds, which we have not done. (3) The cross-model comparison (Experiment 1) does not control for encoder capacity or training budget between F-CS-WAE and the four baselines. (4) Per-class style MMD is a supervised, label-guided regularizer; we do not report a component-wise ablation of F-CS-WAE beyond the  $\delta$  axis (e.g., removing the semantic variable, the class MMD term, or varying the prior geometry), since we do not have completed runs for those variants. (5) Generation self-accuracy is confounded by dataset-intrinsic classification difficulty and should not be used alone as a cross-dataset leakage-severity proxy (Experiment 4). (6) The class-conditional style prior’s failure to fully transfer to CIFAR-10 (Experiment 5) is measured on one checkpoint per dataset; we have not tested whether a richer conditional (e.g., a per-class mixture rather than a single Gaussian) closes the gap. (7) The capacity/curriculum/prior/ supervision robustness check (Experiment 7) perturbs one variable at a time from a single seed-0 checkpoint on each dataset; we have not tested joint perturbations (e.g. small style dimension *and* no classifier together), which could plausibly compound rather than simply add given how much larger each individual effect already is on CIFAR-10 than MNIST, and we have not repeated any perturbation across seeds – so we cannot rule out that some of the effect-size differences between perturbations on CIFAR-10 (e.g. no-warmup’s near-zero effect vs. the auxiliary classifier’s 19.8-point effect) reflect single-run noise rather than a stable ordering.

## 8 Conclusion

Conditional style leakage is a failure mode in factorized generative models: global aggregate regularization of a style variable does not prevent it from becoming class-dependent,

because marginal matching is compatible with  $z_s$  being maximally informative about the class label, not just weakly correlated with it (Proposition 1). We measured this gap with a diagnostic across five model families and showed it is statistically real even when not visually obvious. A controlled latent-swap intervention shows the decoder does not merely have access to the leaked information but is measurably more sensitive to it than to the semantic code (off-diagonal content-following rises from 0.1% to 60.9% on MNIST once the remedy is applied). Perturbing latent capacity, training curriculum, prior geometry, and supervision one at a time on both MNIST and CIFAR-10, no single perturbation eliminates the leakage, so it is not attributable to any one of these choices alone – though on CIFAR-10 its magnitude is highly sensitive to capacity (LP falls by up to 42.9 points, vs. at most 7.4 on MNIST), a sensitivity a MNIST-only check would have missed. Per-class style MMD cuts inter-class style separation by 67–77% at the originally-measured  $\delta=1$  and, across a fuller six-point  $\delta$  sweep, reduces it smoothly on both datasets; the clustering cost is dataset-dependent and, at a single seed per  $\delta$ , noisier than the original two-point comparison suggested – small across most of the sweep on both datasets, with one large outlier per dataset we cannot yet attribute to  $\delta$  itself rather than seed variance. Per-class style MMD also raises MNIST naive-sampling self-accuracy from 15% to 70% (100% combined with the post-hoc style prior). Critically, the remedy that fully repairs MNIST generation only partially repairs CIFAR-10 generation, so fixing the diagnostic finding on one dataset does not automatically fix the underlying generative gap on another. Factorized generative models should verify class-invariance of style variables via conditional diagnostics, not marginal statistics alone, and should report where proposed remedies do and do not generalize rather than presenting a single-dataset result as general.

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## A Architecture Details

The semantic head produces  $(\tilde{\mu}_c, r_c)$ , giving  $\mu_c = \tilde{\mu}_c / \|\tilde{\mu}_c\|_2$  and concentration  $\rho_c = \sigma(r_c)(1 - \varepsilon)$ ; the semantic latent

is sampled by the Spherical Cauchy Möbius reparameterization  $\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1})$ ,  $z_c = T_{\mu_c, \rho_c}(\eta_c)$ . The style head produces  $(\mu_s, \log \sigma_s^2)$  and  $z_s = \mu_s + \sigma_s \odot \varepsilon_s$ ,  $\varepsilon_s \sim \mathcal{N}(0, I)$ . The ResNet-18 backbone for  $32 \times 32$  inputs replaces the first  $7 \times 7$  convolution (stride 2) with a  $3 \times 3$  convolution (stride 1, padding 1) and removes the max-pool layer, yielding a  $4 \times 4$  spatial output with 512 channels; a shared FC layer maps  $512 \times 4 \times 4 = 8,192$  to 256 dimensions with SiLU, from which separate linear heads produce the semantic and style parameters. The decoder concatenates  $[z_c; z_s]$  ( $d_c=64$ ,  $d_s=128$  in all experiments) and maps it through three ResBlockUp stages (bilinear  $2 \times$  up-sample,  $1 \times 1$  projection, GroupNorm+SiLU+Conv residual block;  $512 \rightarrow 256 \rightarrow 128 \rightarrow 64$  channels,  $4 \times 4 \rightarrow 32 \times 32$ ), followed by a  $3 \times 3$  output convolution and sigmoid. For each class  $k$ , F-CS-WAE maintains a center  $m_k \in \mathbb{S}^{d_c-1}$  with prior  $p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p)$ ,  $\rho_p=0.7$ , updated by EMA from batch class-mean directions once per epoch,  $m_k \leftarrow \text{normalize}(\tau m_k + (1 - \tau)\bar{\mu}_c^{(k)})$ ,  $\tau=0.95$ . The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} + \delta(t)\mathcal{L}_{\text{style-cls}} + \eta(t)\mathcal{L}_{\text{cls}}, \quad (4)$$

with  $\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x})$ ,  $\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_k \text{MMD}^2(Q_k, P_k)$  with  $Q_k = \{z_c^i: y_i=k\}$ ,  $P_k \sim p(z_c | y=k)$ ,  $\mathcal{L}_{\text{agg}} = \text{MMD}^2(\{z_c^i\}, \{z_{c,r_j}^p\})$  against a random-class prior mixture, and  $\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y)$  a linear auxiliary classifier preventing semantic collapse.

## B Training Hyperparameters

**Table 8:** Hyperparameters used in all F-CS-WAE experiments.

| Parameter                               | Value                            |
|-----------------------------------------|----------------------------------|
| Optimizer                               | Adam, lr = $10^{-3}$             |
| LR schedule                             | StepLR, step 100, $\gamma = 0.5$ |
| Batch size                              | 128                              |
| Total epochs                            | 300                              |
| Gradient clip (global norm)             | 1.0                              |
| Semantic dim $d_c$ / style dim $d_s$    | 64 / 128                         |
| Prior concentration $\rho_p$            | 0.7                              |
| EMA momentum $\tau$                     | 0.95                             |
| $\lambda_1 / \lambda_{\text{LPIPS}}$    | 1.0 / 0.1                        |
| $\alpha_f / \beta_f$ (class / agg. MMD) | 2.0 / 5.0                        |
| $\gamma_f$ (style)                      | 1.0                              |
| $\delta_f$ (per-class style, off/on)    | 0 / 1                            |
| $\eta_f$ (classifier)                   | 0.3                              |
| Style prior temperature $\tau_s$        | 0.25                             |

Baselines (ResNetAE, VAE, WAE-MMD, VaDE, CS-WAE) use a unified lightweight CNN protocol, 50 epochs, Adam lr  $10^{-3}$ , StepLR(30, 0.5); the cross-model diagnostic baselines of Experiment 1 use a separate lightweight 3-layer

CNN encoder, 100 epochs (see Section 6.1 for the resulting comparability caveat).

## C Phase Schedule Details

**Table 9:** Training phase boundaries and active loss terms; ramps are linear.

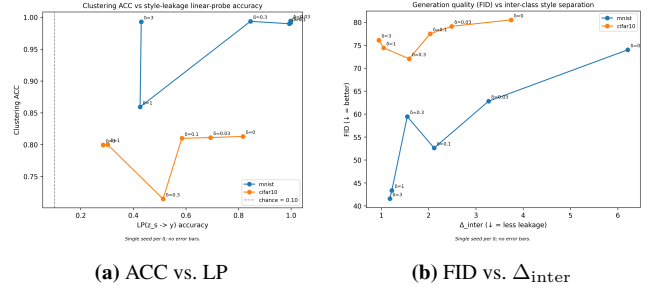
| Phase | Epochs  | Terms and ramp                                                                                                                                                     |
|-------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | 0–49    | $\mathcal{L}_{\text{rec}}$ only                                                                                                                                    |
| B     | 50–99   | $+\mathcal{L}_{\text{style}}(0 \rightarrow \gamma_f)$ , $\mathcal{L}_{\text{style-cls}}(0 \rightarrow \delta_f)$ , $\mathcal{L}_{\text{cls}}(0.1 \rightarrow 0.2)$ |
| C     | 100–199 | $+\mathcal{L}_{\text{class}}(0 \rightarrow \alpha_f/2)$ , $\mathcal{L}_{\text{agg}}(0 \rightarrow \beta_f/2)$ , $\mathcal{L}_{\text{cls}}(0.2 \rightarrow 0.3)$    |
| D     | 200–299 | $\mathcal{L}_{\text{class}}(\alpha_f/2 \rightarrow \alpha_f)$ , $\mathcal{L}_{\text{agg}}(\beta_f/2 \rightarrow \beta_f)$                                          |

Per-class style regularization ramps up in Phase B, before  $\mathcal{L}_{\text{class}}$  and  $\mathcal{L}_{\text{agg}}$  are introduced in Phase C, so that per-class style regularization begins as early as possible relative to the semantic MMD terms introduced later, limiting – rather than eliminating – the epochs during which class-specific  $(z_c, z_s)$  interactions can form without any style-conditional counter-pressure. (Phase A itself, epochs 0–49, is reconstruction-only, so the decoder does see real  $(z_c, z_s)$  pairs before any style or class regularization is active.)

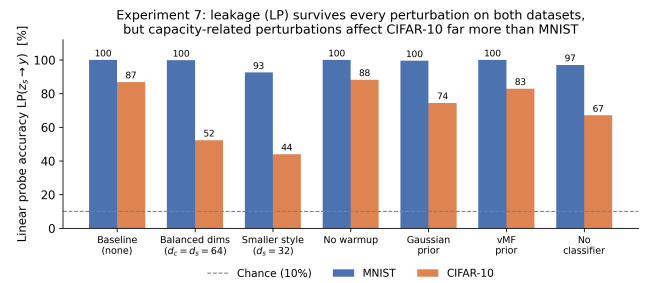
## D t-SNE of the Style Latent Across Model Families



**Figure 9:** CIFAR-10 naive (odd rows) vs. class-conditional  $\tau=0.25$  (even rows) samples, four classes. Diversity within a row collapses under  $\tau=0.25$ , and object identity is often not clearly recognizable by eye even where the auxiliary classifier scores it correctly.



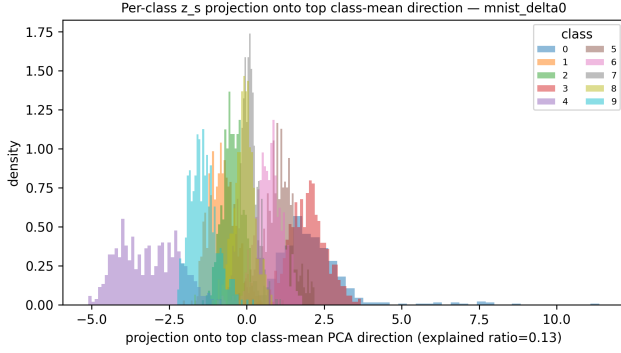
**Figure 10:** Six-point  $\delta$  sweep ( $\delta \in \{0, 0.03, 0.1, 0.3, 1, 3\}$ ), single seed per point, no error bars. Points are annotated with  $\delta$  and connected in increasing- $\delta$  order per dataset; the connecting line is a visual aid, not a claim of a smooth underlying curve (see text). Leakage metrics (LP,  $\Delta_{\text{inter}}$ ,  $x$ -axis of both panels) move smoothly with  $\delta$ ; clustering ACC does not.



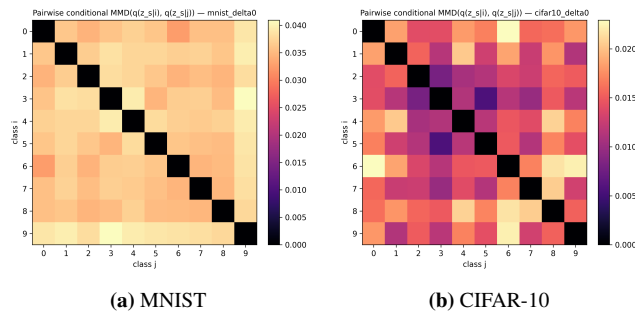
**Figure 11:** Experiment 7: LP( $z_s \rightarrow y$ ) for the baseline and six perturbations, MNIST vs. CIFAR-10. Leakage survives every perturbation on both datasets (dashed line: chance), but the two capacity perturbations (balanced dims, smaller style) move CIFAR-10 far more than they move MNIST, while no-warmup and vMF-prior barely move either.



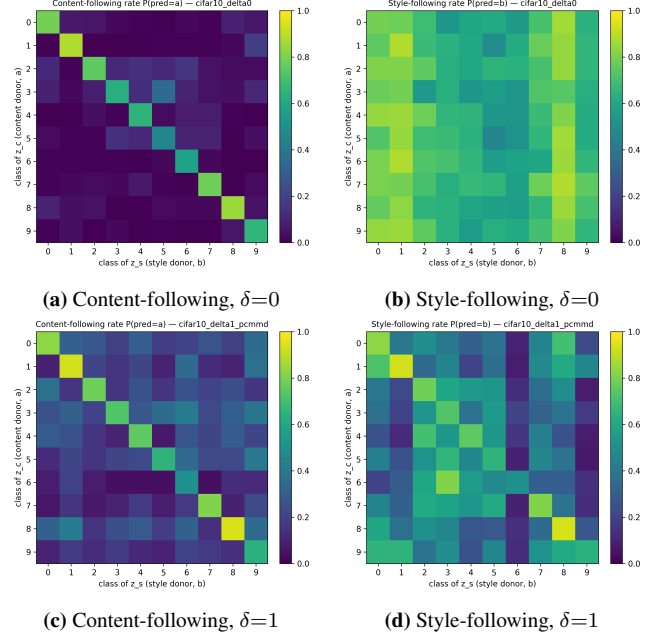
**Figure 12:** t-SNE of the style latent  $z_s$ , colored by MNIST digit class, for all six rows of Table 2. Unlike a typical clustering figure, *color-separated clusters here mean worse* (class information leaking into  $z_s$ ): separation is visually obvious only for WAE-MMD and F-CS-WAE (no per-class MMD)—the two rows with the *largest*  $\Delta_{\text{inter}}$  in Table 2—and visibly reduced (improved) after adding per-class style MMD. For VAE,  $\beta$ -TCVAE, and FactorVAE the leakage is statistically real (LP 74–87%) but not visually apparent in this 2D projection. Referenced from Experiment 1 (Section 6.1).



**Figure 13:** An alternative to the t-SNE view above, for F-CS-WAE at  $\delta=0$ : project  $z_s$  onto the top principal component of the  $K \times d_s$  class-mean matrix (the same signal  $\Delta_{\text{inter}}$  summarizes into one scalar) and plot per-class histograms. Some classes separate cleanly along this single direction even where a 2D t-SNE embedding does not make the separation visually obvious, illustrating that a fixed low-dimensional projection can either overstate or understate leakage depending on which directions it happens to preserve – another reason to prefer the quantitative diagnostics of Section 3.5 over any single plot.

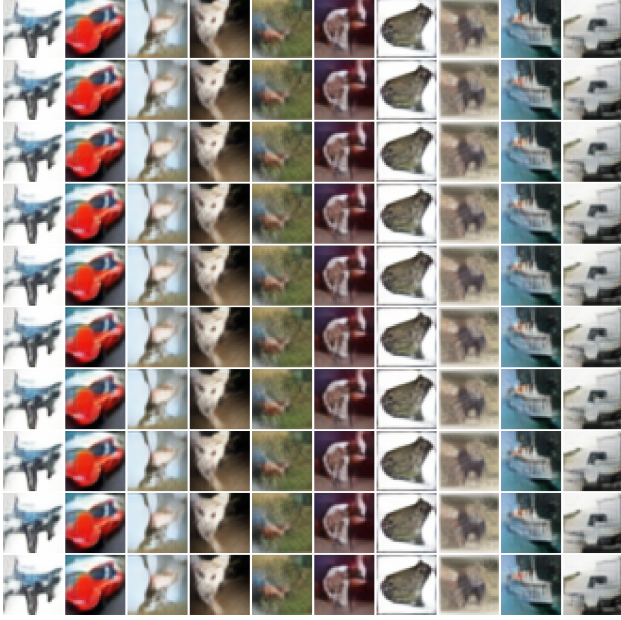


**Figure 14:** Pairwise  $\text{MMD}^2(q(z_s | y=i), q(z_s | y=j))$  between all class pairs, F-CS-WAE at  $\delta=0$ . Off-diagonal structure shows which class pairs occupy the most visibly different regions of  $z_s$ ; both datasets show broad off-diagonal separation, not a few isolated confusable pairs, consistent with the per-class bar chart of Figure 5 showing every class contributes to the leakage rather than a small subset.

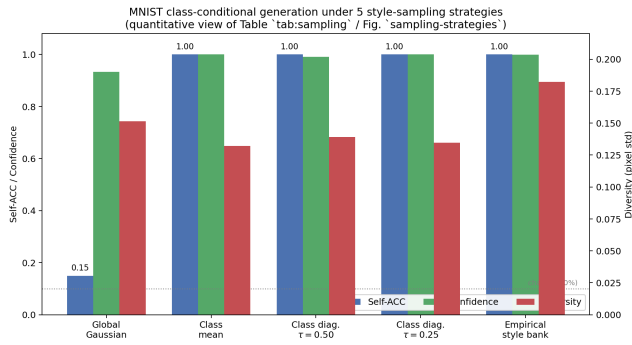


**Figure 15:** CIFAR-10 counterpart to Figure 6. The  $\delta=0 \rightarrow \delta=1$  shift is the same direction as MNIST but weaker in magnitude (content-following 4.2%  $\rightarrow$  20.0%, style-following 68.8%  $\rightarrow$  38.5%), consistent with Section 6.3’s reading that CIFAR-10’s decoder was never as exclusively dependent on  $z_s$  for identity to begin with.

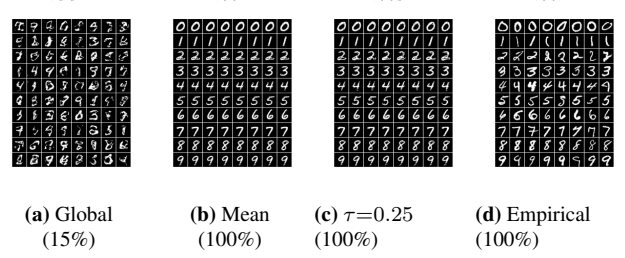
Latent swap: row =  $z_c$  class (content), col =  $z_s$  class (style) — cifar10\_delta0



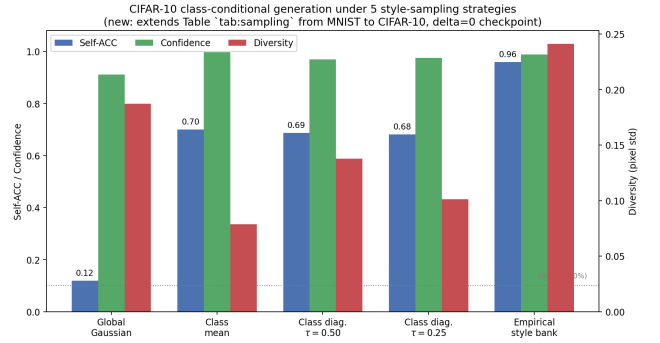
**Figure 16:** CIFAR-10 latent-swap grid,  $\delta=0$  (cf. Figure 7 for MNIST). Rows are visually more similar to each other than in the MNIST grid but still show column-driven (style-driven) drift in object identity for several classes.



**Figure 17:** MNIST sampling strategies (Table 4) plotted directly: self-accuracy jumps from near-chance to 100% the moment  $z_s$  is conditioned on class in any way, while diversity (right axis) is lowest for the class-mean point estimate and highest for the empirical style bank.



**Figure 18:** MNIST generation under representation-aware style sampling. Global Gaussian produces shapeless or wrong-class images; all class-conditional strategies restore correct digit identity.



**Figure 19:** CIFAR-10 analogue of Figure 17 (Table 5). The class-conditional Gaussian strategies that reach 100% on MNIST plateau near 70% here; only resampling real encoded style codes (empirical bank) approaches MNIST-level self-accuracy.